Ratio and Proportion

Number · Edexcel IGCSE Mathematics A (4MA1) · Year 10 · worksheet

A+ Maths Hub

Name	Class	Date

Answer ALL questions.

Write your answers in the spaces provided. You must write down all the stages in your working.

Suggested time: 134 minutes · Total marks: 134

KEY RULES

Simplify	Divide every part by the highest common factor; same units first
Share in a ratio	Add the parts, divide the amount by the total parts, multiply by each part
Unitary method	Find the value of 1, then multiply: cost of 1 = total ÷ number
Best buy	Find the price per 1 kg (or per 1 unit) for each and compare
Combining ratios	Make the common quantity equal: $a : b = 2 : 3$ and $b : c = 4 : 5$
Fractions	Ratio 3 : 2 means $\frac{3}{5}$ and $\frac{2}{5}$ of the total

WORKED EXAMPLE 1

[3 marks]

Share \$240 **in the ratio** 3 : 5 : 4**.**

Step 1	Add the parts	$3 + 5 + 4 = 12$
Step 2	Find the value of 1 part	$240 \div 12 = 20$
Step 3	Multiply each part by 20	60, 100, 80
Step 4	Check: the total is 240	$60 + 100 + 80 = 240$

Answer: \$60, \$100, \$80

WORKED EXAMPLE 2

[3 marks]

Shop A: 750 g of rice costs \$2.40**. Shop B: 1.2 kg of rice costs** \$3.60**. Which is the better buy?**

Step 1	Change to the same unit (kg)	$750\ \text{g} = 0.75\ \text{kg}$
Step 2	Price per kg at Shop A	$2.40 \div 0.75 = 3.20$
Step 3	Price per kg at Shop B	$3.60 \div 1.2 = 3.00$
Step 4	Compare	$3.00 < 3.20$

Answer: **Shop B is cheaper per kg**

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WORKED EXAMPLE 3

[3 marks]

$a : b = 2 : 3$ **and** $b : c = 4 : 5$**. Find** $a : b : c$**.**

Step 1	Make b the same in both ratios (LCM of 3 and 4 is 12)	$b = 12$
Step 2	Multiply $a : b$ by 4	$8 : 12$
Step 3	Multiply $b : c$ by 3	$12 : 15$
Step 4	Join the ratios	$a : b : c = 8 : 12 : 15$

Answer: $8 : 12 : 15$

1 1

1 euro = 120 yen
£1 = 1.2 euros

Change £50 to yen.

..... yen

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · January 2015 · Question 1

2 Green paint can be made by mixing yellow paint and blue paint in the ratio 2 : 3
Wendy makes 15 litres of green paint.

Work out how many litres of blue paint Wendy uses.

..... litres

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · June 2013 · Question 2

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3 A plane has a length of 73 metres.

A scale model is made of the plane.
The scale of the model is 1 : 200

Work out the length of the scale model.
Give your answer in centimetres.

cm

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · January 2019 · Question 1

4 Last season, the number of goals scored by Arjun, by Simon and by Kath for their football team were in the ratios 2 : 5 : 8

Simon scored 12 more goals than Arjun.

Work out the number of goals scored by Kath.

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2023 · Question 1

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5 1 The ocean liner Queen Mary 2 is the longest of its type.
It has a length of 345 metres.



A scale model is made of the Queen Mary 2
The scale of the model is 1 : 200

Work out the length of the scale model.
Give your answer in centimetres.

..... cm

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · June 2015 · Question 1

6 Work out the difference between the largest share and the smallest share when 3450 yen is divided in the ratios 2 : 6 : 7

..... yen

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2018 · Question 2

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7 2 Danil, Gabriel and Hadley share some money in the ratios 3 : 5 : 9
The difference between the amount of money that Gabriel receives and the amount of money that Hadley receives is 196 euros.

Work out the amount of money that Danil receives.

..... euros

(Total for Question 2 is 3 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · January 2022 · Question 2

8 Pritam, Sarah and Emily share some money in the ratios 3 : 6 : 4
Sarah gets $15 more than Emily.

Work out the amount of money that Pritam gets.

$ .....

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · June 2015 · Question 2

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9 The weekly rent for a holiday apartment is £530, which is the same as 715.5 euros.
The weekly rent for a holiday cottage is £750

Using the same rate of currency exchange, work out the weekly rent for the cottage in euros.

..... euros

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · January 2017 · Question 3

10 Yuan sells fudge in small bags and in large bags.

Small bag	**Large bag**
Weight 150 g	Weight 400 g
Cost £1.80	Cost £5

Work out which bag is the better value for money.

Show your working clearly.

Edexcel IGCSE Mathematics A (4MA1) · Paper 2F · November 2023 · Question 15

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11 Anna and Lionel share \$675 in the ratio 4 : 5

Lionel gives $\frac{3}{5}$ of his share of the money to his mother.

How much money does Lionel give to his mother?

\$.....

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2018 · Question 4

12 In a box, there are only green sweets, orange sweets and yellow sweets.

There are 280 sweets in the box so that

the number of green sweets : the number of orange sweets = 2 : 3

and

the number of orange sweets : the number of yellow sweets = 1 : 5

Work out how many green sweets there are in the box.

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · January 2023 · Question 5

13 **5** C grams of chocolate is shared in the ratios 2 : 5 : 8
The difference between the largest share and the smallest share is 390 grams.

Work out the value of C

C = ...

(Total for Question 5 is 3 marks)

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14 A football team played 55 games.
Each game was won, drawn or lost.

number of games won : number of games drawn : number of games lost = 6 : 3 : 2

Work out how many more games the team won than the team lost.

.....

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2019 · Question 7

15 The perimeter of a triangle is 90 cm.
The lengths of the sides of the triangle are in the ratios 3 : 5 : 7

Work out the length of the longest side of the triangle.

..... cm

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · June 2011 · Question 7

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16 Lisa, Max and Punita share £240 in the ratio 3 : 4 : 8

How much more money than Lisa does Punita get?

£.....

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · January 2015 · Question 7

17 Rachel, Mario and Sanjit share some money in the ratios 4 : 3 : 9
Mario receives £96
Work out the difference between the amount received by Rachel and the amount received by Sanjit.

£.....

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · January 2017 · Question 7

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18 The scale of a map is 1 : 25 000
On the map, the distance between two railway stations is 22 cm.

Work out the real distance between the two railway stations.
Give your answer in kilometres.

..... km

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · June 2013 · Question 10

19 The lengths of the sides of a triangle are in the ratios 2 : 6 : 7
The length of the longest side of the triangle is 24.5 cm.

Work out the perimeter of the triangle.

..... cm

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · January 2016 · Question 10

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20 3 years ago, the ratio of Tom's age to Clemmie's age was 2 : 7
Tom is now 15 years old and Clemmie is now x years old.

Find the value of x.

$x =$ .....

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2019 · Question 11

21 Jack, Kate and Lila share some money in the ratios 5 : 9 : 6
In total, Jack and Kate receive £56

Work out the amount of money Lila receives.

£.....

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · January 2013 · Question 11

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22 Here is a list of the ingredients needed to make leek and potato soup for 6 people.

Leek and Potato Soup
Ingredients for 6 people
900 ml chicken stock 900 ml water 750 g leeks 350 g potatoes 350 g onions

(a) Ainsley wants to make leek and potato soup for 13 people.

Work out the amount of chicken stock he needs.

..... ml

(2)

(b) Delia makes leek and potato soup for a group of people.
She uses 1250 g of leeks.

Work out the number of people in the group.

.....

(2)

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · January 2014 · Question 1

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23 Here is a list of the ingredients needed to make lentil soup for 6 people.

Lentil Soup (for 6 people)
120 g lentils 300 g carrots 800 ml vegetable stock 3 onions

Jenny wants to make lentil soup for 24 people.

(a) Work out the amount of vegetable stock she needs.

..... ml

(2)

Ravi is going to make lentil soup.
He uses 450 g of carrots.

(b) How many people is Ravi making the lentil soup for?

.....

(2)

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · January 2016 · Question 1

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24 Here are the ingredients needed to make 12 muffins.

Ingredients to make 12 muffins
300 g flour
150 g sugar
250 ml milk
100 g butter
2 eggs

Sarah makes 60 muffins.

(a) Work out how much sugar she uses.

..... g

(2)

James makes some muffins.
He uses 625 ml of milk.

(b) How many muffins did he make?

.....

(2)

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · June 2016 · Question 1

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25 Here is a list of ingredients needed to make apple and blackberry crumble for 4 people.

Apple and Blackberry Crumble
Ingredients for 4 people

120 grams	flour
80 grams	sugar
90 grams	butter
300 grams	apples
115 grams	blackberries

Rufus wants to make apple and blackberry crumble for 10 people.

(a) Work out the amount of apples he needs.

..... grams
(2)

Roland makes apple and blackberry crumble for a group of people.
He uses 920 grams of blackberries.

(b) Work out the number of people in the group.

.....
(2)

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · June 2017 · Question 3

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26 Nancy has some coins with a total value of 85 pence.
She has only 2 pence coins and 5 pence coins.
The ratio

number of 2 pence coins : number of 5 pence coins = 1 : 3

Nancy has more 5 pence coins than 2 pence coins.

How many more?

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2023 · Question 5

27 **6** Eli and Peta share \$275 in the ratio 2 : 3

Eli gives $\frac{3}{11}$ of his share to charity.

Peta gives 0.32 of her share to charity.

Work out the total amount that Eli and Peta give to charity.

\$.....

(Total for Question 6 is 4 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2025 · Question 6

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28 (a) The length of an Airbus A300 aeroplane is 54 m.
The ratio of the length of this aeroplane to its wingspan is 6 : 5

Work out the wingspan of the aeroplane.

..... m
(2)

(b) A model is made of the Airbus A300 aeroplane.
The length of the model is 36 cm.
The length of the real aeroplane is 54 m.

Find the ratio of the length of the model to the length of the real aeroplane.
Give your ratio in the form 1 : n

1 : .....
(3)

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · June 2012 · Question 1

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29 There are some people in a cinema.

$\frac{3}{5}$ of the people in the cinema are children.

For the children in the cinema,

number of girls : number of boys = 2 : 7

There are 170 girls in the cinema.

Work out the number of adults in the cinema.

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30 Lyn went on holiday to India.
She changed £250 into rupees.

The exchange rate was £1 = 97 rupees.

(a) How many rupees did Lyn get?

..... rupees
(2)

When she returns from holiday, Lyn has **four** 500 rupee notes.
She changes this money into pounds.

The exchange rate is now £1 = 93.5 rupees.

(b) Work out how many pounds Lyn gets.
Give your answer to the nearest pound.

£ .....
(3)

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · June 2017 · Question 3

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31 (a) There are 32 students in a class.
All the students are either left-handed or right-handed.
The ratio of the number of left-handed students to the number of right-handed students is 1 : 7

Work out the number of right-handed students.

.....

(2)

(b) Sajid makes a scale model of a lorry.
He uses a scale of 1 : 32
The length of Sajid's model lorry is 45 cm.
Chitra makes a scale model of the same lorry.
She uses a scale of 1 : 72

Work out the length of Chitra's model lorry.

..... cm

(3)

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · January 2012 · Question 6

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32 7 John changes £450 to euros.

The exchange rate is £1 = 1.16 euros.

(a) Change £450 to euros.

..... euros

(2)

When in Amsterdam, John uses his credit card to pay for a ring costing 850 euros.

He has to pay a bank charge of £3.50 for using his credit card in addition to the cost of the ring.

(b) Work out the total cost, in pounds (£), of the ring and the bank charge.

£ .....

(3)

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · June 2015 · Question 7

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CHALLENGE QUESTIONS

33 A school has 840 pupils and 40 teachers.

(a) Find the ratio of the number of pupils to the number of teachers.
Give your ratio in the form $n : 1$

..... : 1
(2)

In Year 11 at the school, the ratio of the number of pupils who study Chemistry to the number of pupils who study Physics is 3 : 2

(b) 105 pupils in Year 11 study Chemistry.
Work out the number of pupils in Year 11 who study Physics.

.....
(2)

For the 105 pupils who study Chemistry, the ratio of the number of boys to the number of girls is 4 : 3

(c) Work out the number of girls in Year 11 who study Chemistry.

.....
(2)

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · January 2014 · Question 1

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34 On a farm there are chickens, ducks and pigs.

The ratio of the number of chickens to the number of ducks is 7 : 2
The ratio of the number of ducks to the number of pigs is 5 : 9
There are 36 pigs on the farm.

Work out the number of chickens on the farm.

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2021 · Question 4

35 **5** Avril bakes a cake.

She uses flour, butter and sugar such that

weight of flour : weight of butter = 6 : 5
weight of butter : weight of sugar = 3 : 2

Avril uses 120 grams of sugar.

Work out the weight of flour Avril uses.

..... grams

(Total for Question 5 is 3 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2023 · Question 5

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36 Andreas, Isla and Paulo share some money in the ratios 3 : 2 : 5

The **total** amount of money that Isla and Paulo receive is £76 more than the amount of money that Andreas receives.

Andreas buys a video game for £48.50 with some of his share of the money.

Work out how much money Andreas has left from his share of the money when he has bought the video game.

£.....

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2020 · Question 4

37 **6** Pau, Sam and Tia share £240 in the ratios 3 : 4 : 5

Sam and Tia each give £10 of their share to Pau.

Work out the ratios of the amounts of money that Pau, Sam and Tia now have.
Give your answer in its simplest form.

..... : ..... : .....

(Total for Question 6 is 4 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2025 · Question 6

38 Astrid wants to buy some oil.
She can buy the oil from either Dane Oil or Arctic Oil.

Here is information about the price that each company will charge Astrid.

Dane Oil	Arctic Oil
(4.2×10^5) litres for 2 500 000 Krone	(8.6×10^5) litres for 770 000 Dollars

Astrid wants to get the better value for money for the oil.

1 Dollar = 6.57 Krone

TOTAL FOR PAPER IS 134 MARKS

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